TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME ( )

CIVICS GROUP

H2 PHYSICS 9478/03

Paper 3 Longer Structured Questions **23 September 2026**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of the page.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.
The use of an approved scientific calculator is expected, where appropriate.

Section A
Answer **all** questions.

Section B
Answer **one** question only.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		Percentage
Subtotal P1	/ 30	/ 15
Subtotal P2	/ 75	/ 30
Paper 3		
1	/ 7	
2	/ 8	
3	/ 9	
4	/ 9	
5	/ 13	
6	/ 9	
7	/ 20	
8	/ 20	
Deduction		
Subtotal P3	/ 75	/ 35
Subtotal P4	/ 50	/ 20
Grand total		/ 100

This document consists of **24** printed pages and **2** blank pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_o	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on/by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T /K	=	T /°C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean kinetic energy of a molecule of an ideal gas	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_o \sin \omega t$

velocity of particle in s.h.m.	v	=	$v_o \cos \omega t$
		=	$\pm \omega \sqrt{x_o^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
Discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	X	=	$x_o \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 n I$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** questions in this section in the spaces provided.

1 A cyclist is testing a new aerodynamic helmet by measuring the drag force experienced while riding at a steady speed. The drag force is measured using a force sensor attached to the helmet.

The drag force is given by the equation

$$F = \frac{1}{2} k \rho A v^2$$

where k is a unitless constant, ρ is the air density, A is the cross-sectional area, and v is the speed of the cyclist.

(a) Using SI base units, show that the equation is homogenous.

[3]

(b) In an experiment, the cyclist measures the quantity of Ft, where t is the time taken for the experiment. The cyclist obtains slightly different measurements each time.

State two possible random errors in this context.

1……

2. ……

…….. [2]

(c) The force sensor has a zero error.

State whether this error affects the accuracy or the precision of the measurements and suggest how the error can be corrected.

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…….. [2]

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2 **(a)** State Newton's first law of motion.

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……..……..…….

……..…… [1]

(b) Block X of mass 2.0 kg and block Y of mass 4.0 kg, initially at rest, are connected by a light inextensible string that passes over two frictionless light pulleys P_1 and P_2, with one end of the string fixed to the ceiling as shown in Fig. 2.1.

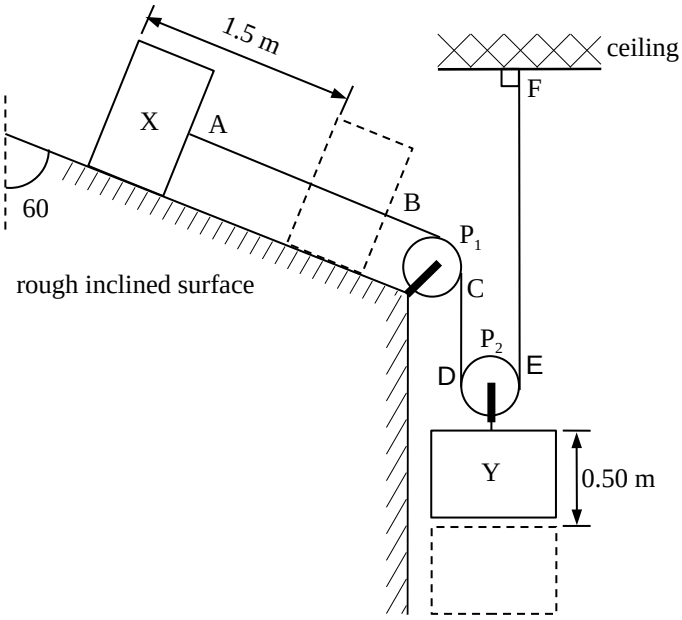


Fig. 2.1

AB is parallel to the inclined surface and CD is parallel to EF.

Both blocks subsequently start to move. The inclined surface exerts a constant frictional force of 5.0 N on block X such that the acceleration of block X is twice the acceleration of block Y.

(i) Calculate the work done by the frictional force when block X moves 1.5 m along the inclined surface.

work done = …… J [1]

(ii) Hence or otherwise, determine the total kinetic energy of the system when block X moves 1.5 m along the inclined surface.

kinetic energy = …… J [2]

(iii) Fig. 2.2 shows a body of system consisting of the string, pulley P_2 and block Y.

On Fig. 2.2, draw and label the forces acting on the body.

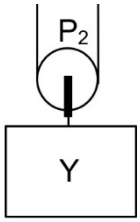


Fig. 2.2

[1]

(iv) Calculate the tension *T* in the string.

T = …… N [3]

3 **(a)** State the principle of conservation of momentum.

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…… [1]

(b) State the relationship between impulse and the force-time graph for a body.

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…… [1]

(c) A ball of mass 0.20 kg was initially moving to the right on a smooth floor with a velocity of 5.0 m s^{-1} before a rightward force *F* was applied to the ball. Fig. 3.1 shows the variation with time of the force *F* applied to an object.

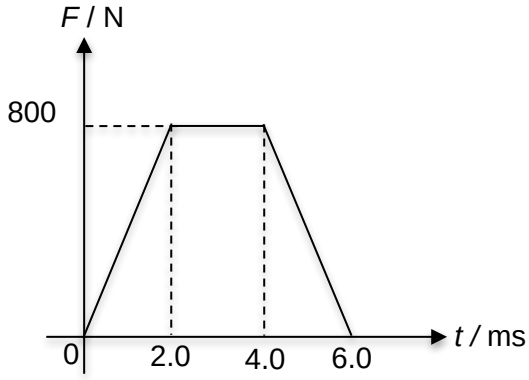


Fig. 3.1

(i) Determine the final momentum of the ball.

momentum = ..…….. N s [3]

(ii) Taking rightwards as positive, sketch the variation with time of the momentum *p* of the ball on Fig. 3.2.

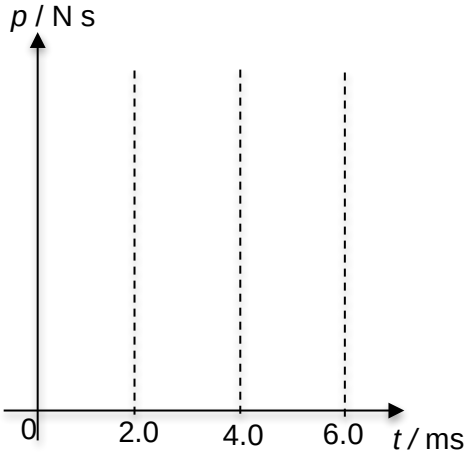


Fig. 3.2

[2]

(d) The ball from part **(c)** now collides with a wall.

With reference to the principle of conservation of momentum, state and explain whether momentum of the **ball** is conserved during collision.

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……...…
. [2]

4 **(a)** Define gravitational field strength at a point.

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…….. [1]

(b) State the relationship between gravitational field strength and gravitational potential at a point.

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…… [1]

(c) A satellite of mass 2200 kg at position P orbits the Earth in a circular path of orbital radius 6.70×10^6 m. as shown in Fig. 4.1.

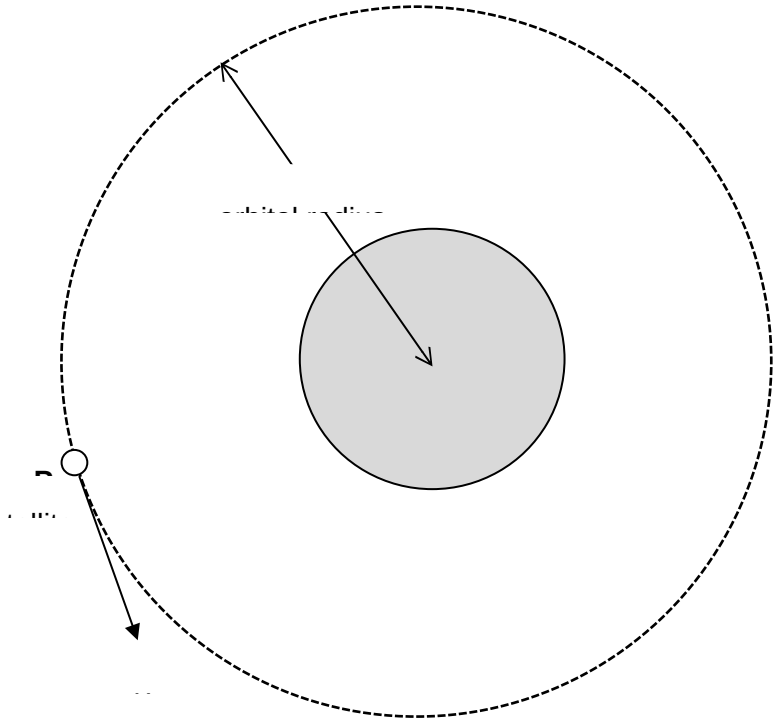


Fig. 4.1

The mass of the Earth is 5.98×10^{24} kg and the radius of the Earth is 6.40×10^6 m.

At point P,

(i) calculate the magnitude of the gravitational field strength g_P,

g_P = …… N kg^{-1} [2]

(ii) calculate the magnitude of the gravitational force on the satellite,

force = …… N [1]

(iii) state the rate of change of potential energy with orbital radius for the satellite,

rate of change of potential energy = …… J m^{-1} [1]

(iv) determine the total energy E_T of the satellite.

E_T = …… J [3]

5 **(a)** Monochromatic light is incident on four rectangular slits as shown in Fig. 5.1.

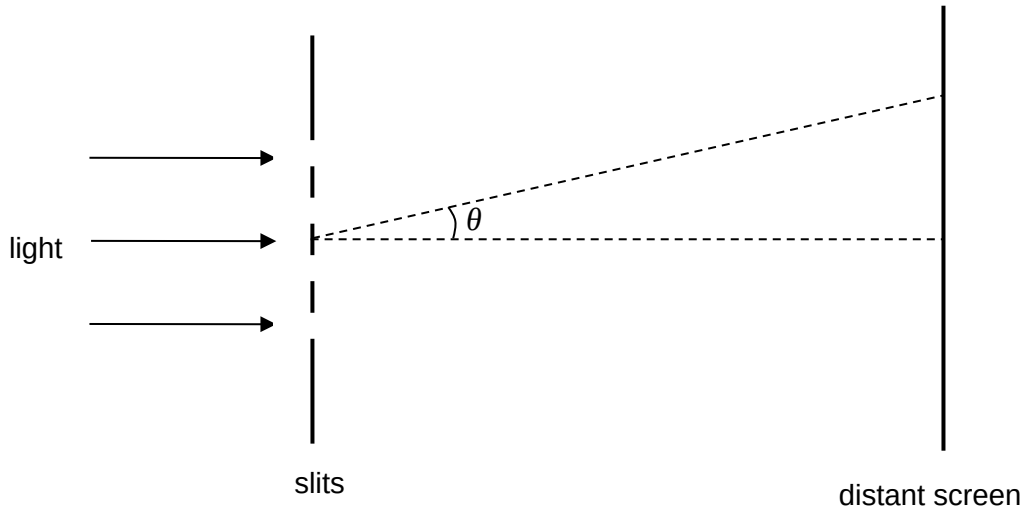


Fig. 5.1 (not to scale)

The graph in Fig. 5.2 shows the variation with angle θ of the intensity of the light I at the distant screen.

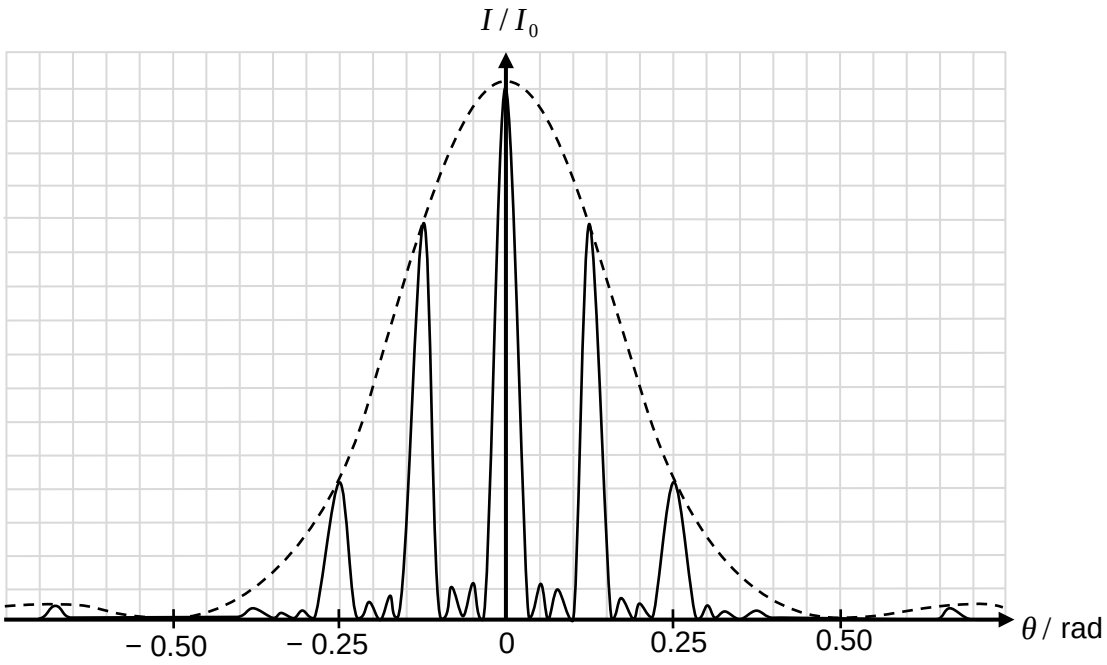


Fig. 5.2

(i) Explain the significance of the dotted curve on the graph.

……

……

…… [1]

(ii) The width of each slit is 1.0 μm.
Use Fig. 5.2 to show that the wavelength of the light is 450 nm.

[1]

(iii) The distance to the screen is 3.2 m.
Determine the separation between the slits.

separation = …… m [2]

(iv) The number of slits is increased while keeping the width of each slit and the separation of slits the same.

State **two** changes to the maxima observed on the screen as a result of the increased number of slits.

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…… [2]

(b) A tube is filled with water and a speaker is placed above the tube, playing a constant

310 Hz tone, as shown in Fig. 5.3 below.

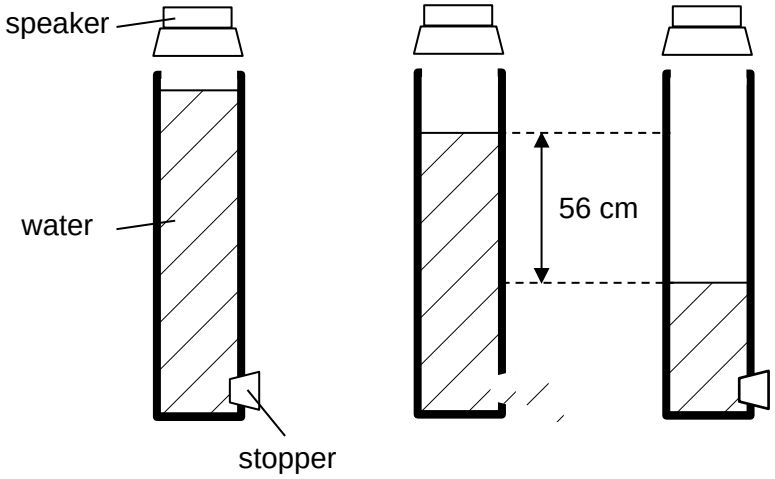


Fig. 5.3 (not to scale)

The stopper is removed and water is allowed to run out of the tube. At the position shown in the centre diagram in Fig. 5.3, a loud sound is heard in the tube for the first time.

Water continues to run out of the tube, and a loud sound is next heard at the position shown in the rightmost diagram in Fig. 5.3.

The stopper is re-inserted to maintain the water level at this position.

(i) On the rightmost diagram of Fig. 5.3, sketch the standing wave in the tube and identify with the letter 'N', the positions of any nodes of the standing wave.

[2]

(ii) Calculate the speed of sound in the tube.

speed of sound = …… m s^{-1} [2]

(iii) With the water level maintained as shown in the rightmost diagram of Fig. 5.3, the frequency of the sound played by the speaker is increased from 310 Hz

until the next loud sound is heard.

Calculate the new frequency of the sound.

frequency = …… Hz [3]

6 **(a)** State what is meant by *mass defect*.

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……..……..

……..… [1]

(b) **(i)** The mass of the helium−3 nuclide $\left({}^{3}_{2}\mathrm{He}\right)$ is 3.014931 u.
The mass of a proton is 1.00728 u while the mass of a neutron is 1.00867 u.

Show that the binding energy per nucleon for the helium−3 nuclide is 2.6 MeV.

[3]

(ii) The binding energy per nucleon for deuterium $\left({}^{2}_{1}\mathrm{H}\right)$ is 1.11 MeV.
Calculate the energy released in the following reaction:

$$ {}^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma $$

energy released = …… MeV [2]

(c) The axes in Fig. 6.1 show the variation with nucleon number *A* with the binding energy

per nucleon.

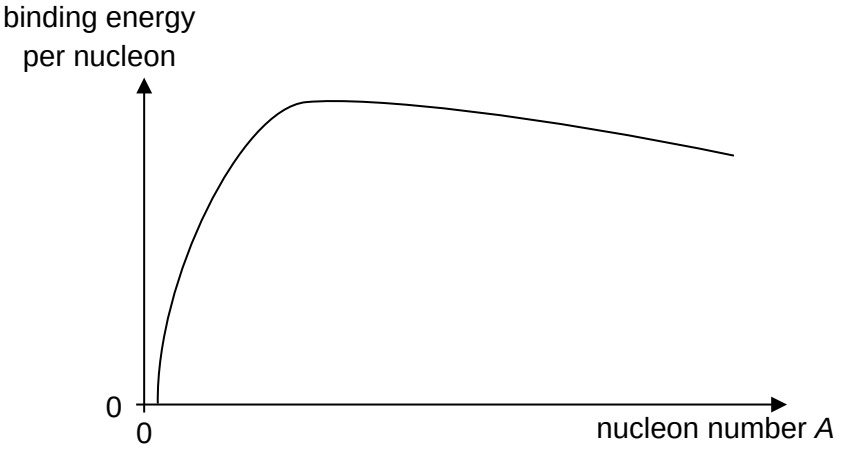


Fig. 6.1

(i) On Fig. 6.1, mark and label two points corresponding to the approximate positions of deuterium $\left({}_{1}^{2}\mathrm{H}\right)$ and helium-3 $\left({}_{2}^{3}\mathrm{He}\right)$.

[1]

(ii) With reference to Fig. 6.1, explain why energy is released in the reaction in **(b) (ii)**.

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…….. [2]

Section B

Answer **one** question from this section in the spaces provided.

7 **(a)** Fig. 7.1 shows a 12.0 V cell with negligible internal resistance connected in series with a uniform resistance wire XY of length 6.0 cm.

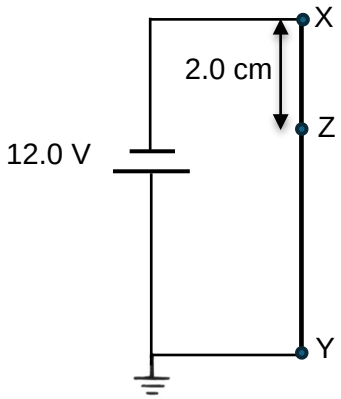


Fig. 7.1

(i) Point Z is 2.0 cm away from point X. Determine the potential at point Z.

potential at point Z = …….…… V [3]

(ii) Show that the resistance of a wire is given by

$$R = \frac{\rho D L^2}{m}$$

where ρ is the resistivity of the wire, D is the density of the wire, L is the length of the wire and m is the mass of the wire.

[2]

(iii) A student suggests that, if the length of wire XY is doubled with same cross sectional area, the resistance of wire XY increases by a factor of four.

Explain whether the student is correct.

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. [2]

(b) **(i)** The *I*-*V* characteristic graph for a NTC thermistor is shown in Fig. 7.2.

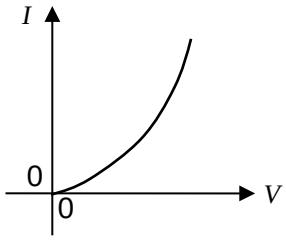


Fig. 7.2

Explain how the resistance of a NTC thermistor changes with increase in potential difference in terms of the number density of charge carriers.

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. [4]

(ii) The NTC thermistor is connected in a circuit with a cell, a resistor and a light emitting diode (LED) so that the LED emits light at high temperature.

Draw a circuit diagram showing how the components are connected and explain how the LED is able to emit light at high temperature.

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. [4]

(iii) Explain why the resistor is needed in the circuit in part **b(ii)**.

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……... [1]

(c) An initially uncharged capacitor of capacitance 2.0 μF is charged by a battery of e.m.f. 5.0 V through a resistor of resistance 3.0 MΩ, as shown in Fig. 7.3. The switch is closed at time $t = 0$.

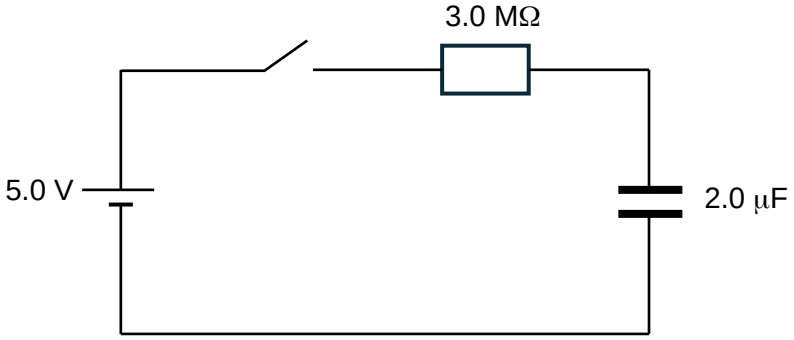


Fig. 7.3

(i) In Fig. 7.4, sketch the variation with time t of potential difference V across the capacitor, until it is fully charged.

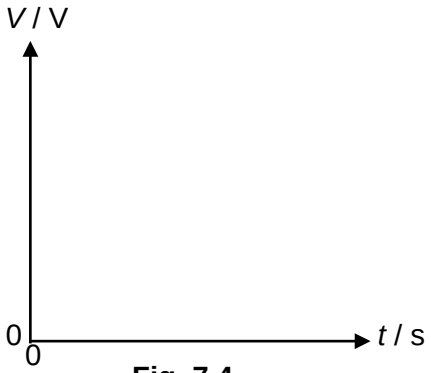


Fig. 7.4

[2]

(ii) Determine the time taken to charge the capacitor to 2.0 V.

time = …… s [2]

8 **(a)** State what is meant by a *field of force*.

……

…… [1]

(b) Fig. 8.1 shows the setup of a high-resolution spectrometer within a vacuum. It is used to measure masses of singly positive charged ions (+*e*) which are released from an ion source.

The ions enter a velocity selector at right angles to both electric (*E*) and magnetic (B_1) fields. The uniform electric field (*E*) is setup with a pair of horizontally charged plates, namely plate P and plate Q. The uniform magnetic flux density B_1 is directed into the plane of the paper.

Only ions with velocity *v* pass through the velocity selector undeflected, before entering into another uniform magnetic field (B_2) region. The ions X and ions Y follow semicircular paths before reaching a detection surface as shown in Fig. 8.1.

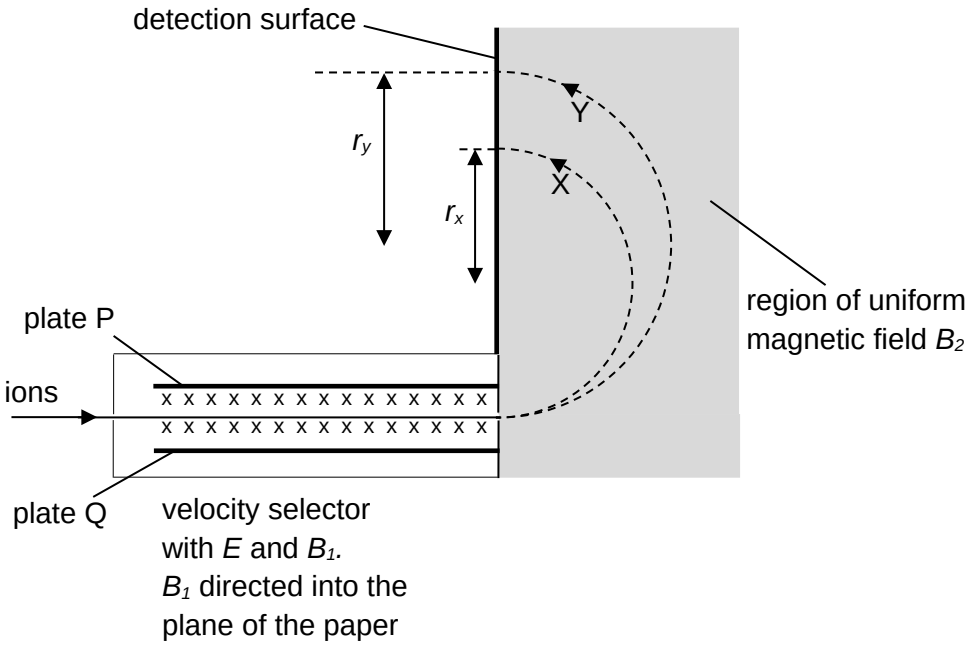


Fig. 8.1

The measured electric field strength, flux densities, radii of the paths of the ions and their associated uncertainties are shown below.

electric field strength *E* / V m^{-1}	3.0×10^4 ±2%
magnetic flux density B_1 / T	0.25 ±1%
magnetic flux density B_2 / T	0.60 ±1%
radius of path of ion *X*, r_x / m	0.074 ±0.001
radius of path of ion *Y*, r_y / m	0.079 ±0.001

(i) Explain why the ions pass undeflected through the velocity selector.

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…… [2]

(ii) State and explain which plate, P or Q is at a higher potential.

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…… [2]

(iii) Calculate the ions' velocity v.

v = …… m s^{-1} [2]

(iv) State and explain the direction of the magnetic flux density B_2.

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…… [2]

(v) Explain whether there is work done on the ion as it moves within B_2.

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…… [2]

(vi) Considering the path of ion within B_2 region, show that the expression for the

ion mass is $m = \frac{eB_1B_2r}{E}$.
where m is the ion mass and r is the radius of the semicircular path by the ion.

Explain your working.

[2]

(vii) Calculate the mass of ion X.

mass of ion X = …… kg [1]

(viii) Calculate the percentage uncertainty of the mass of ion X.

percentage uncertainty = …… % [2]

(ix) The spectrometer resolution criterion is given by the equation

$$\Delta r > 3\sigma$$

where Δr is the difference in the radius of path of the two X and Y ions and σ is the absolute uncertainty of the radius of the path.

Determine whether the ions are distinguishable.

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…… [2]

(x) Suggest and explain one experiment modification to improve resolution without changing E.

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…… [2]

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